Name _____ Surname _____ Date ____/____/_____

Mock test — Speed, energy, work, power and energy resources

Formative test, not graded. Time: 50 minutes. Calculator allowed; formula sheet not allowed. Use $g = 10\,\text{N/kg}$. **This paper is your fair copy:** the letters in Part A, the drawings on the graph and everything you write in the spaces provided must be free of erasures and cross-outs. Do your rough work on a separate sheet and copy only the final version here. Write the equation in symbols before the numbers and give the unit in each answer. Total: 50 marks.

⚽ **Lifebuoy.** The parts marked with the lifebuoy are worth 30 marks out of 50, that is 60%: if you have completed all of them, you are safe. The other parts take you beyond the pass mark.

Part A — Multiple choice and true or false

Exercise 1 — Multiple choice

⚽ Only one answer is correct. Each question is worth 1 mark, 10 marks in total. Write the letter in the box.

1. A car travels 90 km in 1 h 15 min. Its average speed is:
   A 90 km/h B 113 km/h C 72 km/h D 1.2 km/h ☐
2. On a distance–time graph, line P is steeper than line Q. This means that P shows an object that:
   A is moving faster than Q B is further from the start than Q
   C is speeding up D has been moving for longer ☐
3. The speed of a trolley increases from 2.0 m/s to 6.0 m/s. Its kinetic energy increases by a factor of:
   A 3 B 4 C 6 D 9 ☐
4. A 4.0 kg object is lifted by 2.5 m at constant speed. The gain in its gravitational potential energy is:
   A 10 J B 100 J C 40 J D 400 J ☐
5. A ball is thrown vertically upwards. Air resistance is negligible. At the highest point of its flight:
   A its kinetic energy is at its maximum B its total energy is zero
   C its gravitational potential energy is at its maximum D its kinetic energy equals its gravitational potential energy ☐
6. A person pushes a box with a force of 40 N through 3.0 m in 5.0 s. The work done by the person is:
   A 8 J B 120 J C 200 J D 600 J ☐
7. A motor lifts a 20 kg load through 3.0 m in 4.0 s. The useful power of the motor is:
   A 150 W B 600 W C 200 W D 2400 W ☐
8. A machine wastes 30 kJ of energy while it transfers 90 kJ usefully. Its efficiency is:
   A 25% B 33% C 300% D 75% ☐
9. Two machines are connected in a chain. The first is 80% efficient and the second is 50% efficient. The overall efficiency is:
   A 30% B 65% C 40% D 130% ☐
10. Which of these energy resources does not depend on energy from the Sun?
    A wind B geothermal C biofuel D hydroelectric ☐

Exercise 2 — True or false

⚽ Write T or F. No explanation needed. Each statement is worth 1 mark, 4 marks in total.

(a) A horizontal line on a distance–time graph shows an object moving at constant speed. ☐

(b) A power of 1 W is the same as 1 J/s. ☐

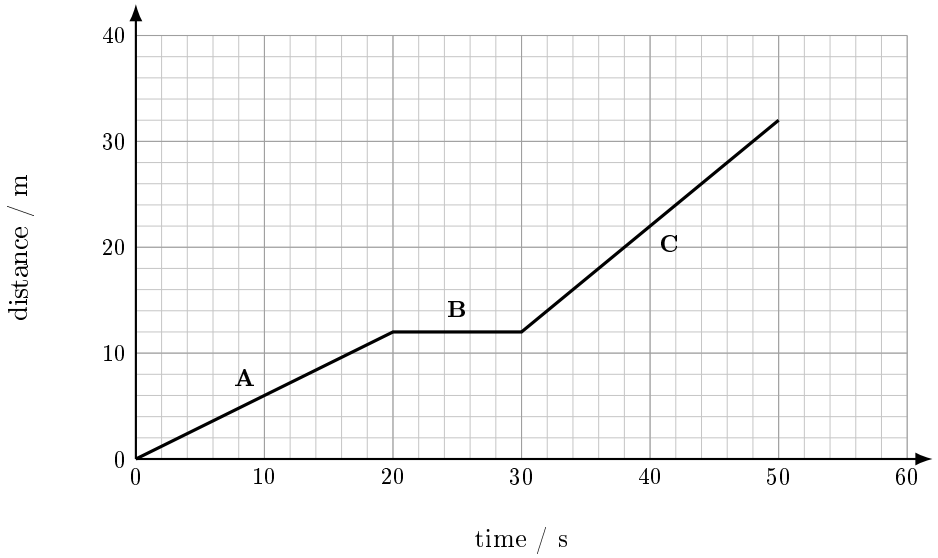
(c) Two objects have the same speed. The heavier one has more kinetic energy. ☐

(d) A machine that is 60% efficient wastes 60% of the energy supplied to it. ☐

Part B — Structured questions

Exercise 3 — A delivery robot

A delivery robot moves along a straight corridor. The graph shows its distance from the kitchen against time. The journey has three sections, A, B and C.



(a) ⚽ Which section shows the robot stationary? Which section shows the greatest speed? Write the letters. **[2]**

stationary: ..... greatest speed: .....

(b) ⚽ State what the gradient of a distance–time graph represents. **[1]**

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(c) Calculate the speed of the robot in section A, in m/s. **[2]**

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Answer .....

(d) Calculate the average speed of the robot for the whole journey, in m/s. **[2]**

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Answer .....

(e) ⚽ A second robot makes the same journey from the kitchen to the same final position at a constant speed, without stopping, and takes the same total time. Draw its line on the graph. **[1]**

Exercise 4 — A stone thrown upwards

A stone of mass 0.20 kg is thrown vertically upwards from the ground with a speed of 12 m/s. Ignore air resistance.

(a) ⚽ Name the energy store that decreases and the energy store that increases while the stone rises. **[1]**

decreases: ..... increases: .....

(b) ⚽ Write, in symbols and without numbers, the equation that gives the maximum height h of the stone in terms of its launch speed u and g. **[2]**

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(c) Calculate the maximum height of the stone. **[2]**

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Answer .....

(d) Calculate the speed of the stone when it is 3.0 m above the ground. **[3]**

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Answer .....

(e) ⚽ The stone is replaced by one of mass 0.40 kg, thrown with the same speed. The maximum height is greater / the same / smaller. Circle one choice and give a one-line reason. **[1]**

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Exercise 5 — Energy resources

(a) ⚽ Complete the table. For each resource write R (renewable) or N (non-renewable) and name the energy store that the energy comes from. 1 mark for each row. **[4]**

Resource	**R or N**	**Energy store**
hydroelectric		
natural gas		
geothermal		
nuclear fission		

(b) ⚽ State one advantage of using wind turbines instead of coal to generate electricity. **[1]**

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Exercise 6 — Pushing a crate

A person pushes a crate of mass 25 kg along a rough horizontal floor with a constant horizontal force of 80 N. The crate moves 6.0 m at constant speed.

(a) Write, in symbols, the equation for the work done by a constant force F that moves an object through a distance d in the direction of the force. **[1]**

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(b) Calculate the work done by the person. **[2]**

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Answer .....

(c) In which energy store does the energy transferred by the person end up? **[1]**

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(d) The same force of 80 N now pushes the crate over a smooth floor with no friction, starting from rest. Calculate the speed of the crate after 6.0 m. **[3]**

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Answer .....

Exercise 7 — An electric hoist

An electric hoist lifts a load of mass 120 kg through a height of 4.0 m in 8.0 s. The input electrical power of the motor is 800 W.

(a) Calculate the useful energy transferred to the load. **[2]**

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Answer .....

(b) Calculate the useful power output of the hoist. **[2]**

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Answer .....

(c) Calculate the efficiency of the hoist, as a percentage. **[2]**

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Answer .....

(d) State where the wasted energy goes. **[1]**

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